

THUNDER BAY, ON, Canada

WMO: 710720

Lat: 48.3694N

Lon: 89.3272W

Elev: 199

StdP: 98.95

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.5	(c) -26.7	(d) -34.4	(e) 0.2	(f) -28.9	(g) -31.7	(h) 0.2	(i) -26.2	(j) 10.3	(k) -12.5	(l) 9.3	(m) -11.8	(n) 2.7	(o) 250	(p) 0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	29.0	20.4	27.2	19.2	25.5	18.3	21.8	27.0	20.5	25.3	19.4	23.8	4.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	15.0	24.9	18.8	13.9	23.2	17.7	13.0	21.7	64.8	27.3	60.1	25.4	56.2	23.9	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.7	(b) 8.5	(c) 7.5	(e) -33.9	(f) 32.3	(g) 3.8	(h) 1.9	(i) -36.7	(j) 33.7	(k) -38.9	(l) 34.9	(m) -41.1	(n) 35.9	(o) -43.9	(p) 37.4		
(a) 9.7	(b) 8.5	(c) 7.5	(e) -34.1	(f) 24.2	(g) 3.8	(h) 1.1	(i) -36.8	(j) 25.0	(k) -39.0	(l) 25.7	(m) -41.2	(n) 26.3	(o) -43.9	(p) 27.1		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-13.4	-11.4	-4.8	2.6	9.2	14.8	18.1	17.3	12.8	5.8	-1.7	-9.2
(7)		DBStd	12.07	7.50	7.03	6.65	4.50	3.92	3.42	3.00	3.14	4.26	4.36	5.74	7.02
(8)		HDD10.0	3182	724	598	458	226	63	3	0	0	20	145	351	594
(9)		HDD18.3	5514	982	831	717	473	285	114	42	58	172	389	601	852
(10)		CDD10.0	784	0	0	1	2	37	148	251	226	104	15	1	0
(11)		CDD18.3	75	0	0	0	0	1	9	34	25	6	0	0	0
(12)		CDH23.3	921	0	0	0	1	39	148	400	274	58	1	0	0
(13)		CDH26.7	202	0	0	0	0	7	32	97	55	11	0	0	0
(14)	Wind	WSAvg	3.3	3.3	3.4	3.4	3.6	3.6	3.2	3.0	2.9	2.9	3.4	3.3	3.3
(15)	Precipitation	PrecAvg	731	33	25	41	52	73	84	86	89	87	64	54	42
(16)		PrecMax	1063	84	63	104	144	180	174	224	174	290	172	118	101
(17)		PrecMin	517	7	4	10	4	9	33	8	27	25	9	10	12
(18)		PrecStd	147	19	17	23	28	36	36	49	38	54	38	29	23
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	3.7	6.1	13.7	21.3	27.5	30.1	31.4	30.5	28.1	22.1	14.5	5.5	
(20)		0.4% MCWB	2.0	3.1	8.2	11.2	15.8	20.6	21.7	22.1	20.0	15.5	9.6	3.0	
(21)		2% DB	1.6	3.0	8.7	16.4	23.5	26.9	29.2	28.2	24.8	18.6	10.3	3.4	
(22)		2% MCWB	0.5	1.1	4.6	8.4	13.8	17.9	20.7	20.4	18.7	13.4	7.0	2.0	
(23)		5% DB	0.2	1.3	5.8	13.5	20.0	24.8	27.4	26.3	22.5	15.7	8.3	2.0	
(24)		5% MCWB	-0.6	0.1	3.0	6.8	12.5	16.9	19.7	19.3	17.4	11.9	6.1	1.1	
(25)		10% DB	-1.9	-0.6	3.7	10.7	17.2	22.6	25.6	24.6	20.4	13.1	6.3	0.8	
(26)		10% MCWB	-2.8	-2.0	1.6	5.4	10.9	15.8	18.7	18.4	16.2	10.3	4.6	-0.1	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	2.2	3.5	9.2	12.0	17.7	22.1	23.8	23.3	21.5	17.0	10.4	3.8	
(28)		0.4% MCDB	3.3	5.6	12.7	19.3	24.5	28.9	29.6	28.9	26.3	20.3	13.2	4.8	
(29)		2% WB	0.7	1.5	5.3	9.3	15.1	19.3	21.9	21.7	19.7	14.3	7.9	2.2	
(30)		2% MCDB	1.4	2.6	7.9	15.2	20.6	24.6	27.3	26.5	23.0	17.6	9.7	3.0	
(31)		5% WB	-0.7	0.1	3.3	7.4	13.5	17.9	20.6	20.4	18.2	12.3	6.2	1.2	
(32)		5% MCDB	0.2	1.5	5.5	12.2	18.5	23.0	25.7	24.9	21.5	14.9	7.8	2.0	
(33)		10% WB	-2.8	-1.9	1.9	5.8	11.9	16.6	19.4	19.2	16.9	10.6	4.6	-0.1	
(34)		10% MCDB	-1.9	-0.7	3.7	10.2	16.4	21.1	24.2	23.2	19.6	12.8	6.2	0.7	
(35)	Mean Daily Temperature Range	MDBR	10.6	11.8	11.4	11.7	12.6	12.3	12.4	12.5	12.0	9.9	8.5	9.1	
(36)		5% DB MCDBR	10.0	11.1	12.4	18.0	18.2	16.4	15.2	15.2	14.7	14.5	11.6	9.0	
(37)		5% WB MCWBR	9.2	9.4	8.9	10.8	10.0	8.8	8.0	8.4	9.0	9.8	8.7	7.9	
(38)		5% WB MCDBR	9.4	10.3	11.0	16.0	15.1	14.0	13.9	13.9	12.6	12.6	10.2	8.3	
(39)	Clear-Sky Solar Irradiance	taub	0.262	0.272	0.306	0.346	0.362	0.382	0.398	0.380	0.357	0.316	0.290	0.265	
(40)		taud	2.399	2.367	2.378	2.369	2.404	2.371	2.373	2.424	2.477	2.557	2.524	2.444	
(41)		Ebn at Noon	825	896	913	905	900	880	859	861	849	838	788	775	
(42)		Edh at Noon	65	84	99	112	113	117	116	105	91	71	59	56	
(43)	All-Sky Solar Radiation	RadAvg	1.08	2.10	3.39	4.72	5.47	5.88	5.93	5.10	3.66	2.16	1.14	0.88	
(44)		RadStd	0.13	0.22	0.29	0.48	0.36	0.41	0.33	0.38	0.26	0.27	0.11	0.09	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page